

ENGR 1411: Material Science

Lecture 2: Crystalline Structures

Upcoming due dates

- Due today: HW1 - Module 1 Prelab Assignment
- Due Tuesday: HW 2 - Module 2 Prelab Worksheet
- Next Class (Tuesday): Module 1 completed in class

We can describe the *crystal* structure for each!



Steel



Crystal



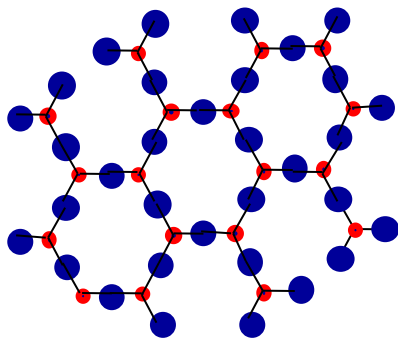
Glass

Crystals → ordered arrangement of atoms, ions, or molecules in a crystalline material

Materials and Packing

Crystalline materials...

- atoms pack in periodic, 3D arrays
- typical of:
 - metals
 - many ceramics
 - some polymers



crystalline SiO₂

Adapted from Fig. 3.40(a),
Callister & Rethwisch 3e.

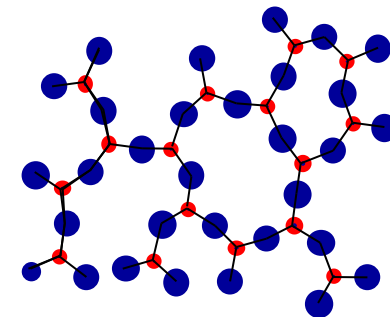
• Si

• Oxygen

Noncrystalline materials...

- atoms have no periodic packing
- occurs for:
 - complex structures
 - rapid cooling

"Amorphous" = Noncrystalline

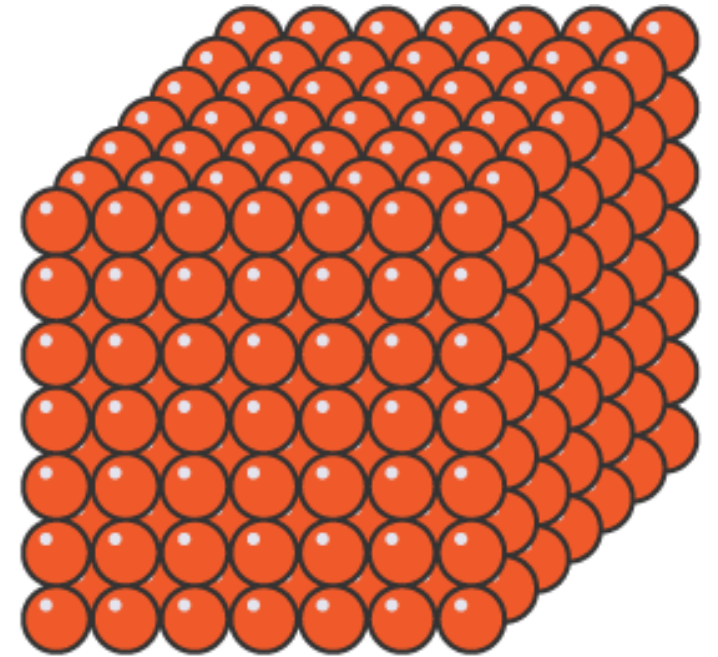


noncrystalline SiO₂

Adapted from Fig. 3.40(b),
Callister & Rethwisch 3e.

Metallic Crystal Structures – simplest crystal structure

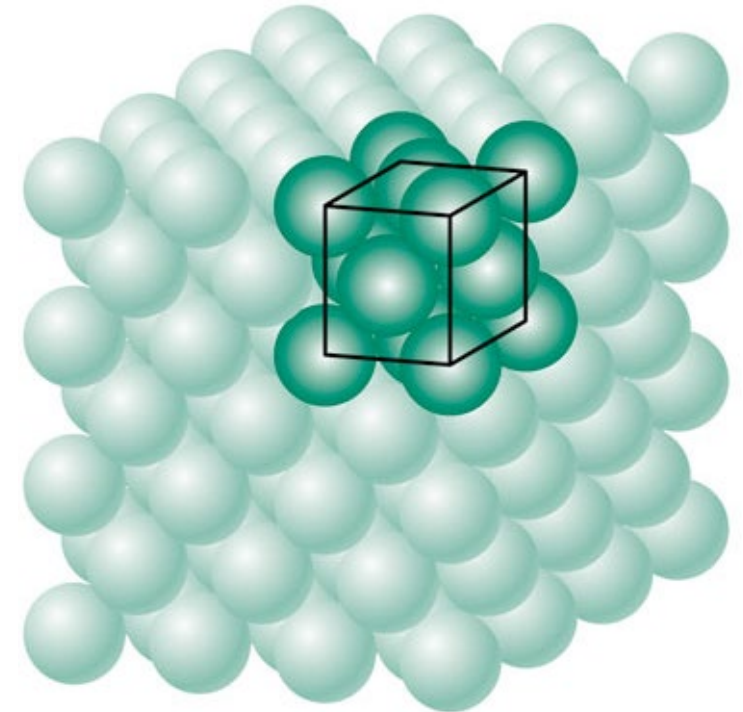
- Dense packing due to
 - All atomic radii are the same
 - Metallic bonding is nondirectional
 - Nearest neighbor distance tends to be small-lower bond energy



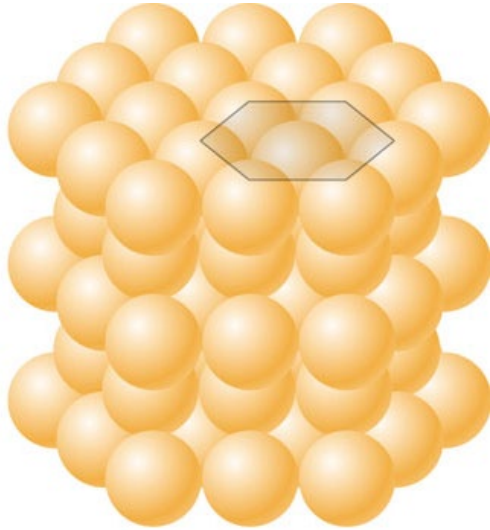
Metallic Crystal Structures

- Unit cell - building block of atoms which repeats itself to form the crystal structure
- Space lattice – imaginary network of lines connecting atoms
- Coordination number – number of nearest neighbor or touching atoms
- Atomic Packing Factor: $APF = \frac{V_{atoms}}{V_{unit\ cell}}$
- R is the radius of an atom

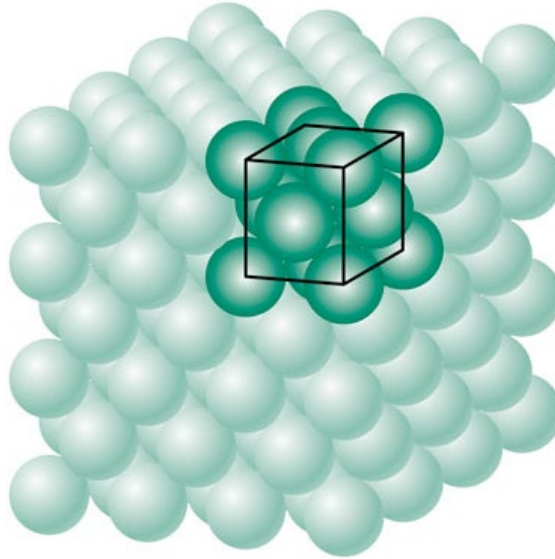
Example:



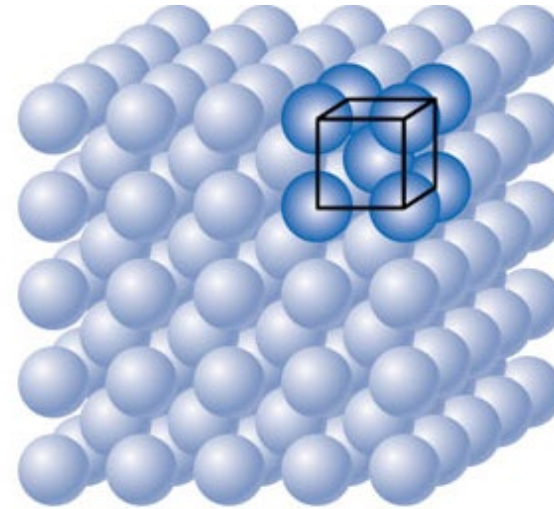
Metallic Crystal Structures



HCP



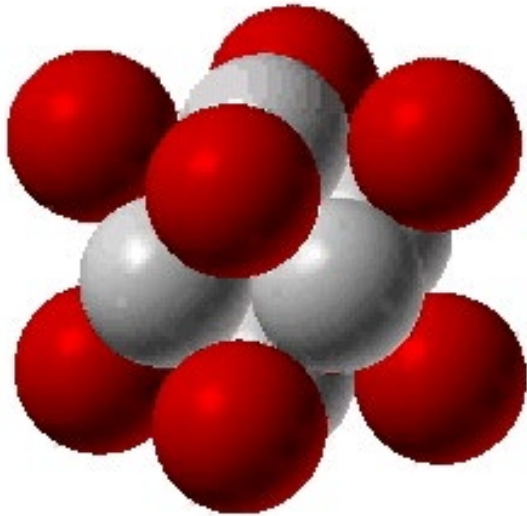
FCC



BCC

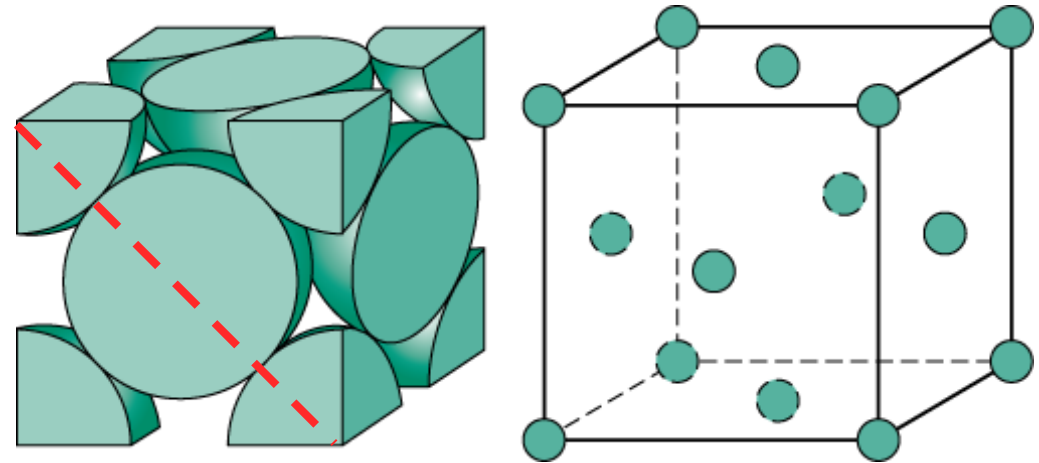
Face Centered Cubic Structure (FCC)

- Atom centered on each face
- Atoms touch each other along face diagonals.



Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

(Courtesy P.M. Anderson)



Adapted from Fig. 3.1, Callister & Rethwisch 3e.

Coordination # = 12

4 atoms/unit cell: $6 \text{ face} \times \frac{1}{2} + 8 \text{ corners} \times \frac{1}{8}$

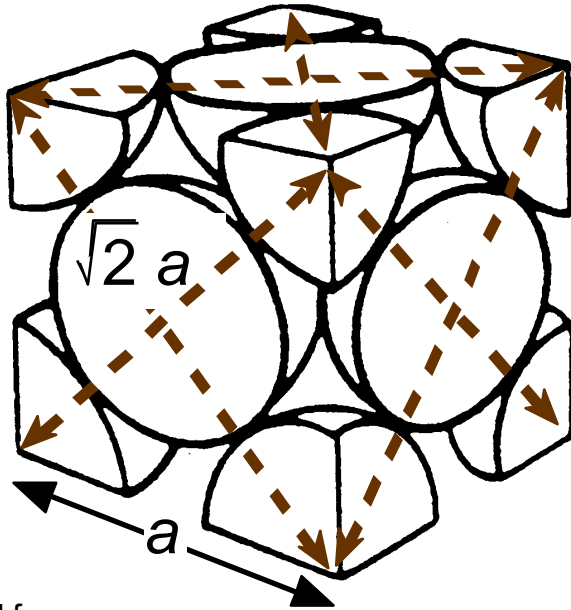
ex: Al, Cu, Au, Pb, Ni, Pt, Ag

FCC Calculations:

For a FCC unit cell, determine the relationship between side length (a) and atomic radius (R)

Using the relationship determined above, what is the atomic packing factor (APF) for the FCC cell?

Atomic Packing Factor: FCC



Close-packed directions:
length = $4R = \sqrt{2} a$

Unit cell contains:
 $6 \times 1/2 + 8 \times 1/8$
= 4 atoms/unit cell

Adapted from
Fig. 3.1(a),
Callister &
Rethwisch 3e.

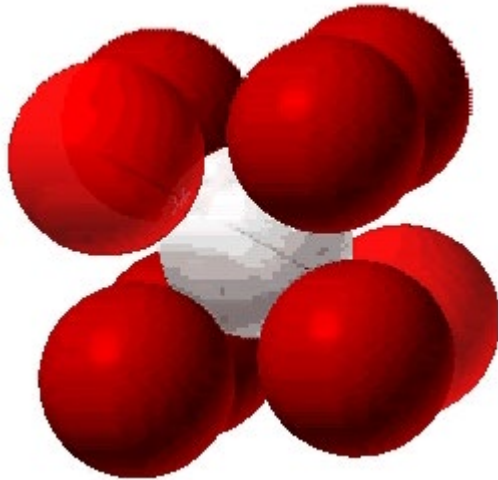
$$\text{APF} = \frac{\frac{\text{atoms}}{\text{unit cell}} \cdot \frac{4}{3} \pi R^3}{a^3}$$

Labels in the diagram:
- $\frac{\text{atoms}}{\text{unit cell}}$ (green text, points to 4)
- $\frac{4}{3} \pi R^3$ (points to the volume of one atom)
- $\frac{\text{volume}}{\text{atom}}$ (brown text, points to $\frac{4}{3} \pi R^3$)
- a^3 (blue text, points to the volume of the unit cell)
- $\frac{\text{volume}}{\text{unit cell}}$ (blue text, points to a^3)

APF for a face-centered cubic structure = 0.74

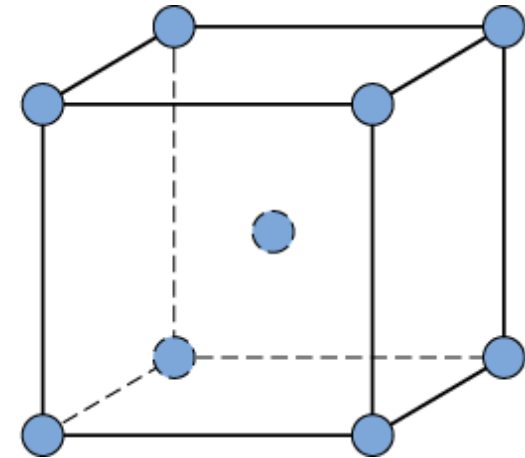
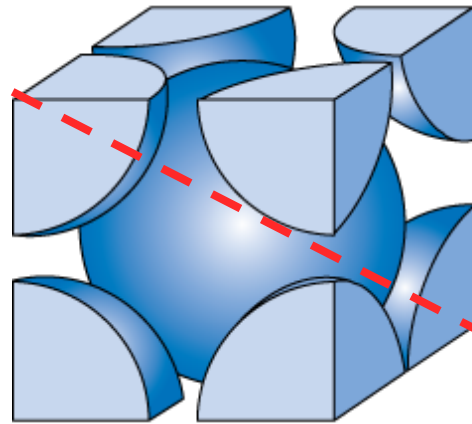
Body Centered Cubic Structure (BCC)

- Body of Atom is centered in cube
- Atoms touch each other along cube diagonals.



Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.

(Courtesy P.M. Anderson)



Adapted from Fig. 3.2,
Callister & Rethwisch 3e.

Coordination # = 8

2 atoms/unit cell: 1 center + 8 corners $\times$ 1/8

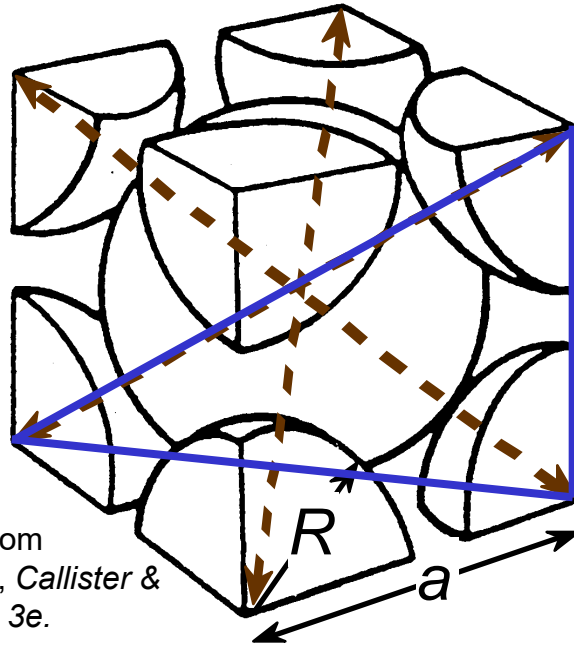
ex: Chromium (Cr), Tungsten (W), Ferrite (Fe- α)

BCC Calculations:

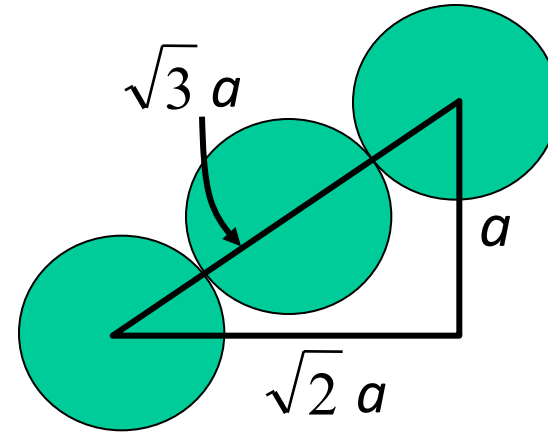
For a BCC unit cell, determine the relationship between side length (a) and atomic radius (R)

Using the relationship determined above, what is the atomic packing factor (APF) for the BCC cell?

Atomic Packing Factor: BCC



Adapted from
Fig. 3.2(a), Callister &
Rethwisch 3e.



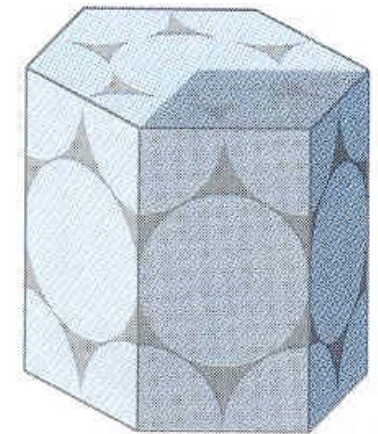
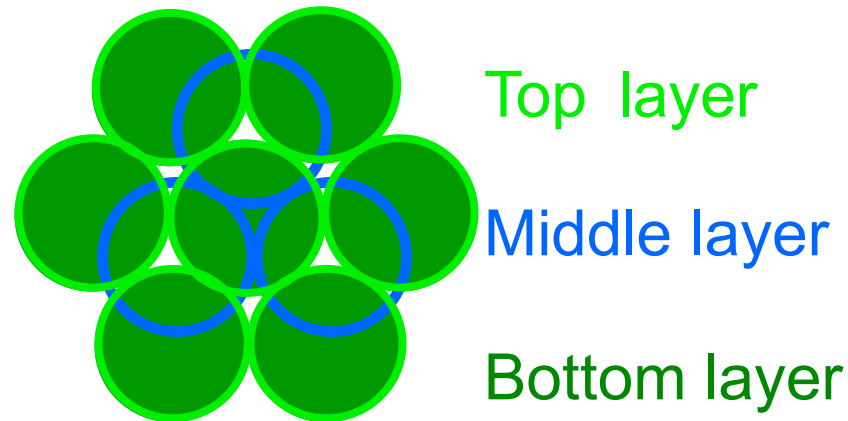
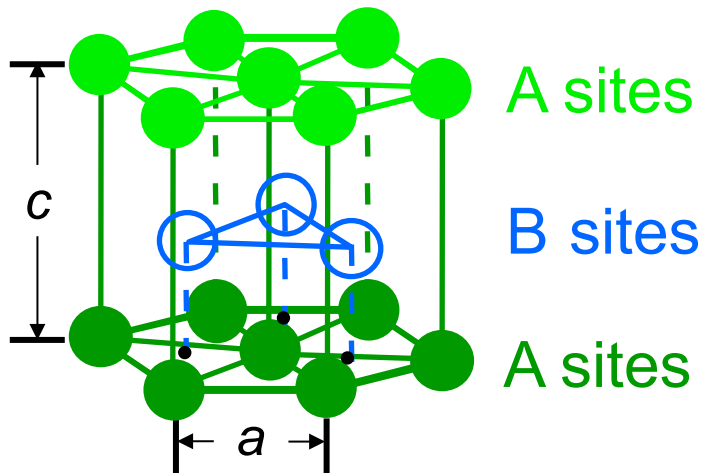
Close-packed directions:
length = $4R = \sqrt{3}a$

$$\text{APF} = \frac{\overbrace{2 \frac{4}{3} \pi R^3}^{\text{atoms}}}{\underbrace{a^3}_{\text{unit cell}}} \quad \leftarrow \frac{\text{volume}}{\text{atom}}$$

APF for a body-centered cubic structure = 0.68

Hexagonal Close-Packed Structure (HCP)

ABAB... Stacking Sequence



Adapted from Fig. 3.3(a),
Callister & Rethwisch 3e.

- Coordination # = 12

6 atoms/unit cell

- APF = 0.74

ex: Cd (cadmium), Mg, Ti, Zn

- $c/a = 1.633$

Reference Videos (to watch on your own)

HCP:

https://www.youtube.com/watch?v=_OhQv5gsoSQ

C to a ratio in HCP: <https://www.youtube.com/watch?v=k4vUoT0LI6o&t=6s>

https://www.youtube.com/watch?v=X-0XX_OYLdg

BCC and FCC:

<https://www.youtube.com/watch?v=uwf8FHIvWKA>

Simple Cubic Unit Cell

<https://www.youtube.com/watch?v=olevxVKgJFg>

Metal Theoretical Density, ρ

$$\text{Density} = \rho = \frac{\text{Mass of Atoms in Unit Cell}}{\text{Total Volume of Unit Cell}}$$

$$\rho = \frac{n A}{V_C N_A}$$

where

n = number of atoms/unit cell

A = atomic weight

V_C = Volume of unit cell = a^3 for cubic

N_A = Avogadro's number
= 6.022×10^{23} atoms/mol

Example:

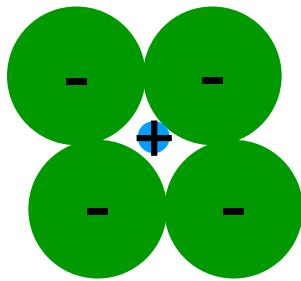
Hafnium (Hf) crystallizes with a body-centered cubic (BCC) structure. It has an Atomic Mass of 178.49 amu (Atomic Weight of 178.49 g/mole) and a density of 13.31 g/cm^3 .

1. Given the above data, calculate the length of the edge of the BCC unit cell
2. Calculate the radius of the Hf atom when bonded into a crystalline solid

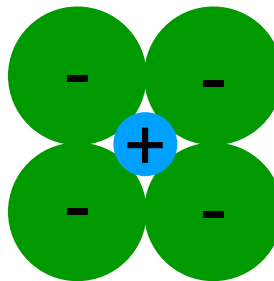
Ceramics - Factors that Determine Crystal Structure

1. Relative sizes of ions – Formation of stable structures:

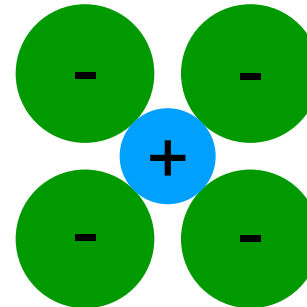
--maximize the # of oppositely charged ion neighbors.



unstable



stable



stable

Adapted from Fig. 3.4,
Callister & Rethwisch 3e.

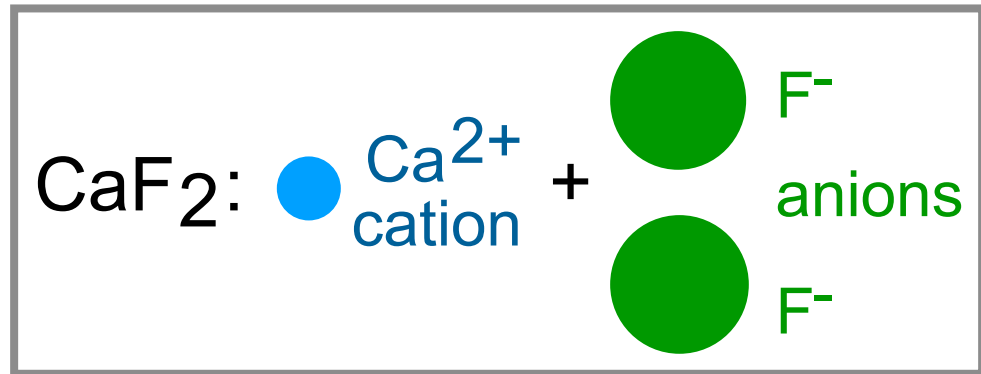
2. Maintenance of Charge Neutrality :

--Net charge in ceramic should be zero.

--Reflected in chemical formula:



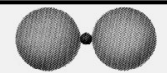
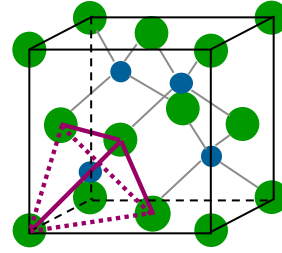
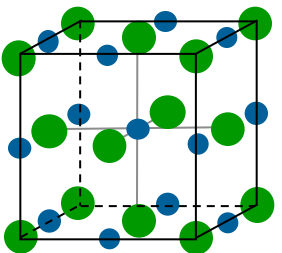
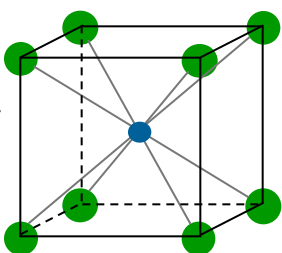
m, p values to achieve charge neutrality



Ceramics - Coordination # and Ionic Radii

Coordination # increases with $\frac{r_{\text{cation}}}{r_{\text{anion}}}$

To form a stable structure, how many anions can surround around a cation?

$\frac{r_{\text{cation}}}{r_{\text{anion}}}$	Coord #			
< 0.155	2	linear		
0.155 - 0.225	3	triangular		
0.225 - 0.414	4	tetrahedral		
0.414 - 0.732	6	octahedral		
0.732 - 1.0	8	cubic		

Adapted from Table 3.3, Callister & Rethwisch 3e.

ZnS (zinc blende)
Adapted from Fig. 3.7, Callister & Rethwisch 3e.

NaCl (sodium chloride)
Adapted from Fig. 3.5, Callister & Rethwisch 3e.

CsCl (cesium chloride)
Adapted from Fig. 3.6, Callister & Rethwisch 3e.

What element is a diamond?



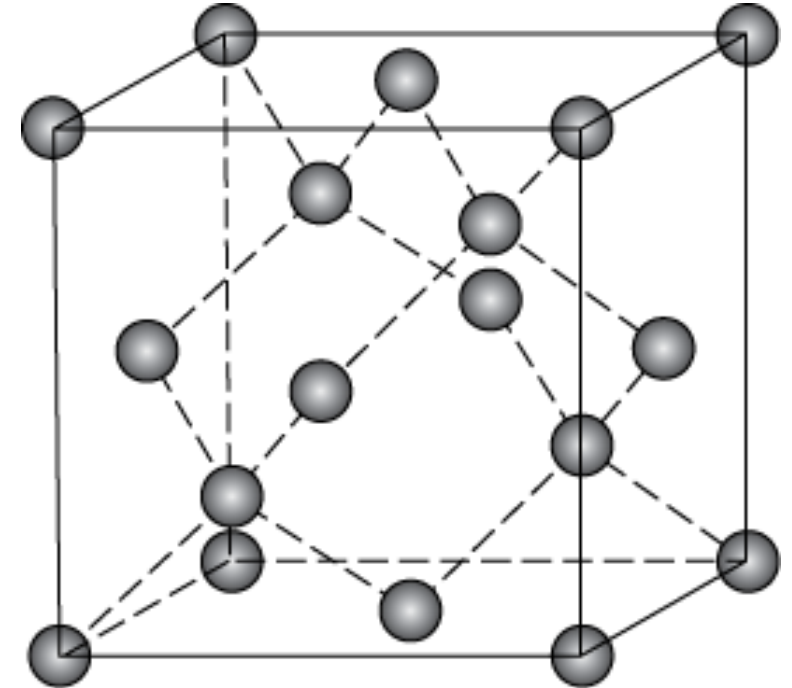
What element is a diamond made up of ?



Polymorphic Forms of Carbon

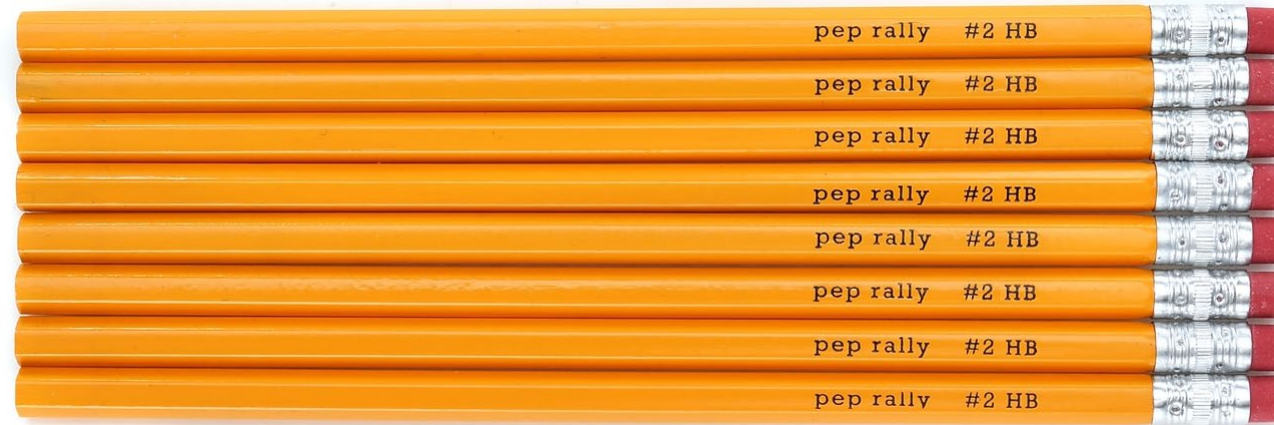
Diamond

- tetrahedral bonding of carbon
 - hardest material known
 - very high thermal conductivity
- large single crystals – gem stones
- small crystals – used to grind/cut other materials
- diamond thin films
 - hard surface coatings – used for cutting tools, medical devices, etc.



Adapted from Fig. 3.16,
Callister & Rethwisch 3e.

What is a pencil lead primarily made of?



There is no lead in pencils. Rather, the core is made up of a non-toxic mineral called graphite. The common name “pencil lead” is due to an historic association with the stylus made of lead in ancient Roman times.

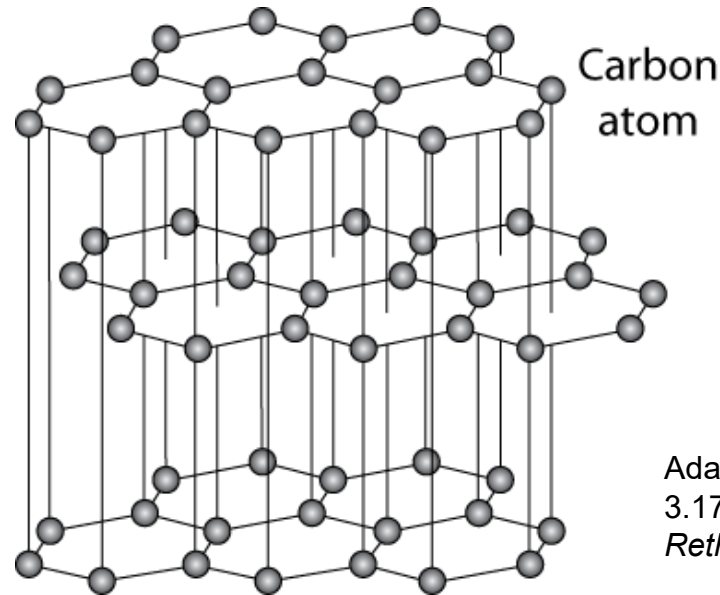
How To Use A Pencil To Open a Sticky Key Lock

- <https://www.youtube.com/watch?v=E7vSuyCnvE8>
- Why?

Polymorphic Forms of Carbon (cont)

Graphite

- layered structure – parallel hexagonal arrays of carbon atoms



Adapted from Fig.
3.17, *Callister &
Rethwisch 3e.*

- weak van der Waal' s forces between layers
- planes slide easily over one another -- good lubricant

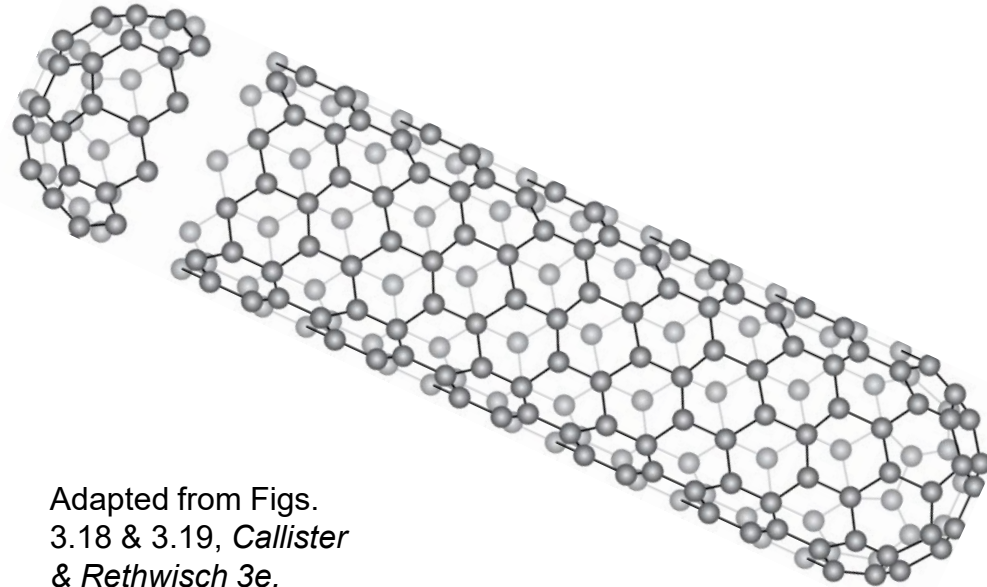
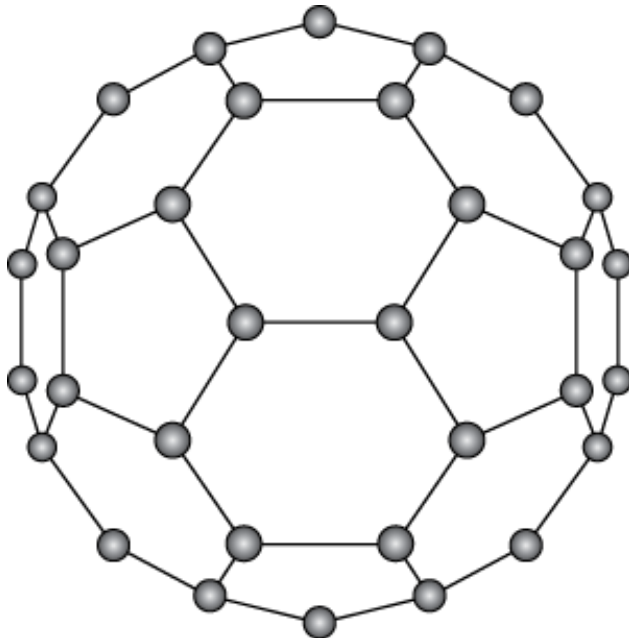
Graphene:

<https://www.youtube.com/watch?v=frgetR7qJec>

Polymorphic Forms of Carbon (cont)

Fullerenes and Nanotubes

- **Fullerenes** – spherical cluster of 60 carbon atoms, C_{60}
 - Like a soccer ball
- **Carbon nanotubes** – sheet of graphite rolled into a tube
 - Ends capped with fullerene hemispheres



Adapted from Figs.
3.18 & 3.19, *Callister
& Rethwisch 3e.*

Module 2 – Signups and Lab

- Dr. Gargac has emailed out instructions.
- Make sure you sign up on the online spreadsheet!